RAJALAKSHMI ENGINEERING COLLEGE

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R2

Started on	Tuesday, 15 September 2026, 12:16 AM
State	Finished
Completed on	Tuesday, 15 September 2026, 12:17 AM
Time taken	43 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	**g**	**t**	**a**	**b**
s2	**g**	x	**t**	x	**a**	y	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <string.h>

#define MAX_LEN 1005

// Helper function to find the maximum of two integers
int max(int a, int b) {
    return (a > b) ? a : b;
}

// Function to find the length of the Longest Common Subsequence
int lcs(char *s1, char *s2) {
    int m = strlen(s1);
    int n = strlen(s2);

    // dp array stores the results of the current row
    int dp[MAX_LEN] = {0};

    // Iterate through characters of the first string
    for (int i = 1; i <= m; i++) {
        int prev = 0; // Stores dp[i-1][j-1] before it gets overwritten

        // Iterate through characters of the second string
        for (int j = 1; j <= n; j++) {
            int temp = dp[j]; // Backup current dp[j] for the next column's prev

            if (s1[i - 1] == s2[j - 1]) {
                dp[j] = prev + 1;
            } else {
                dp[j] = max(dp[j], dp[j - 1]);
            }

            prev = temp; // Update prev to be used in the next iteration
        }
    }

    return dp[n];
}

int main() {
    char s1[MAX_LEN];
    char s2[MAX_LEN];

```

```
    // Read input strings (handles lines containing whitespace up to newline)
    if (scanf("%s", s1) != 1) return 0;
    if (scanf("%s", s2) != 1) return 0;

    // Print the length of the longest common subsequence
    printf("%d\n", lcs(s1, s2));

    return 0;
}
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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